

MT8002 Elements of Set Theory
7.2 Ordered Sets. Linearly Ordered Sets

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Preliminaries

Textbook

Derek Goldrei (1996). Classic Set Theory. A Guided Independent Study.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Outline

Introduction

Binary Relations

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Strict Orders

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Introduction

- We can use binary relations to order **some** (partial order) or **all** (linear order) the elements of a set.
- Orders are binary relations that satisfy certain properties.
- **Weak** orders and **strict** orders can be interchangeably defined.

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Binary Relations

Definition (p. 164)

Let X be a set. A set R is a **binary relation on X** if $R \subseteq X \times X$.

Notation

Let R be a binary relation on a set X .

- (a) We write $(x, y) \in R$ or xRy , if $x \in X$ is related to $y \in X$ by the relation R .
- (b) We write $(x, y) \notin R$ or $x \not R y$, if $x \in X$ is not related to $y \in X$ by the relation R .

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Weak Orders

Definitions (p. 164)

A binary relation R on a set X is a **weak partial order** on the set X (or the set X is **weakly partially ordered** by the relation R), if it has the following properties:

- (a) **reflexive**: for all $x \in X$, xRx ;
- (b) **anti-symmetric**: for all $x, y \in X$, if xRy and yRx then $x \doteq y$;
- (c) **transitive**: for all $x, y, z \in X$, if xRy and yRz then xRz .

Weak Orders

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- (c) **transitive**: for all $x, y, z \in X$, if xRy and yRz then xRz .

If additionally the binary relation R has the following property:

- (d) **linear**: for all $x, y \in X$, xRy or yRx ,
- then the relation R is a **weak linear order** on the set X (or the set X is **weakly linearly ordered** by the relation R).

Weak Orders

Examples (informal, p. 164)

The usual relations $\leq$ and $\geq$ on the sets $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$ are weak linear orders on those sets.

Weak Orders

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Example

Let X be a set. The relation $\subseteq$ on $\mathcal{P}(X)$ is a weak partial order.

Weak Orders

Exercise

Let X be a set. Is the relation $\subseteq$ on $\mathcal{P}(X)$ a weak linear order?

Weak Orders

Exercise

Let X be a set. Is the relation $\subseteq$ on $\mathcal{P}(X)$ a weak linear order?

Answer

No! For example the relation $\subseteq$ on the power set of $\{a, b\}$ is not linear because $\{a\}, \{b\} \in \mathcal{P}(\{a, b\})$ but neither $\{a\} \subseteq \{b\}$ nor $\{b\} \subseteq \{a\}$.

Weak Orders

Description

The **restriction** of a weak partial order to a subset is the weak partial order obtained by discarding all comparisons involving elements outside of the subset.

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Definition (p. 164)

Let R be a weak partial order on a set X and let $A \subseteq X$. The **restriction** of the relation R to the set A , denoted $R|_{A \times A}$, is the relation defined by

$$R|_{A \times A} := \{ (a, b) \in R : a, b \in A \}.$$

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Theorem (Exercise 7.2)

Let R be a weak partial (respectively linear) order on a set X and let $A \subseteq X$. Show that $R|_{A \times A}$ is a partial (respectively linear) order on the set A .

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Strict Orders

Definitions (p. 165)

A binary relation S on a set X is a **strict partial order** on X (or the set X is **strictly partially ordered** by the relation S) if it has the following properties:

- (a) **irreflexive**: for all $x \in X$, $x \not S x$;
- (b) **transitive**: for all $x, y, z \in X$, if $x S y$ and $y S z$ then $x S z$.

Strict Orders

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If additionally the binary relation S has the following property:

- (c) **linear**: for all $x, y \in X$, $x S y$ or $x \doteq y$ or $y S x$,
- then the relation S is a **strict linear order** on the set X (or the set X is **strictly linearly ordered** by the relation S).

Strict Orders

Examples

The usual relations $<$ and $>$ on the sets $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$ are strict linear orders on those sets.

Example

Let X be a set. The relation $\subset$ on $\mathcal{P}(X)$ is a strict partial order.

Strict Orders

Theorem (relationship between weak and strict partial orders, Exercise 7.4)

(a) Let R be a weak partial order on a set X . The binary relation S defined by

$$xSy := xRy \text{ and } x \neq y$$

is a strict partial order on the set X .

Strict Orders

Theorem (relationship between weak and strict partial orders, Exercise 7.4)

(a) Let R be a weak partial order on a set X . The binary relation S defined by

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is a strict partial order on the set X .

(b) Let S be a strict partial order on a set X . The binary relation R defined by

$$xRy := xSy \text{ or } x \doteq y$$

is a weak partial order on the set X .

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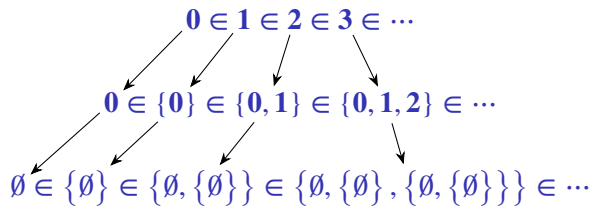
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Orders on Natural Numbers

An “extra” property of natural numbers



Orders on Natural Numbers

Theorem 3.6

The set of natural numbers $\mathbb{N}$ is strictly linearly ordered by the binary relation $\in$.

Orders on Natural Numbers

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The set of natural numbers $\mathbb{N}$ is strictly linearly ordered by the binary relation $\in$.

That is, the binary relation $\in$ on $\mathbb{N}$ has the following properties:

- (a) **irreflexive**: for all $n \in \mathbb{N}$, $n \notin n$;
- (b) **transitive**: for all $m, n, p \in \mathbb{N}$, if $m \in n$ and $n \in p$ then $m \in p$;
- (c) **linear**: for all $m, n \in \mathbb{N}$, $m \in n$ or $m \doteq n$ or $n \in m$.

Orders on Natural Numbers

Example (p. 42)

Although the binary relation $\in$ is transitive on the set $\mathbb{N}$ it does not mean that the relation is transitive on any set. For example, the relation is not transitive on the set

$$\{0, \{0\}, \{\{0\}\}\}$$

because $0 \in \{0\}$ and $\{0\} \in \{\{0\}\}$ but $0 \notin \{\{0\}\}$.

Orders on Natural Numbers

Definition (p. 41)

For all $\mathbf{m}, \mathbf{n} \in \mathbb{N}$,

$\mathbf{m} < \mathbf{n} := \mathbf{m} \in \mathbf{n}$,

$\mathbf{m} \leq \mathbf{n} := \mathbf{m} < \mathbf{n}$ or $\mathbf{m} \doteq \mathbf{n}$.

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Order-Isomorphism

Conventions

- “We shall often write the order relation on X as $<_x$ or $\leq_x$, depending on whether the order is strict or weak.” (p. 166)
- Sometimes a (weakly or strictly; partially or linearly) ordered set X by a relation R will we denote by the pair (X, R) .[†]

[†]Note that the textbook does not use this convention.

Order-Isomorphism

Motivation

We need a precise way to express when two ordered sets are “the same”.

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Definitions (p. 166)

Let $<_X$ and $<_Y$ be strict partial orders on the sets X and Y , respectively.

The sets X and Y are **order-isomorphic**, denoted $X \cong Y$ or $(X, <_X) \cong (Y, <_Y)$, if there is a **bijection** $f : X \longrightarrow Y$ such that for all $x_1, x_2 \in X$,

$$x_1 <_X x_2 \quad \text{if and only if} \quad f(x_1) <_Y f(x_2).$$

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The map f is an **order-isomorphism**.

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The map f is an **order-isomorphism**.

Remark

The above definition on weak partial orders is similar.

Order-Isomorphism

Exercise (informal)

Let $<$ and $>$ be the usual orders on $\mathbb{Z}$. Show that $(\mathbb{Z}, <) \cong (\mathbb{Z}, >)$.

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Solution

Let $f : \mathbb{Z} \longrightarrow \mathbb{Z}$ be the function $n \longmapsto -n$. We can show that the function f is a bijection and that it preserves the order, that is, for all $m, n \in \mathbb{Z}$,

$$m < n \quad \text{if and only if} \quad -m > -n.$$

Order-Isomorphism

Theorem (Exercise 7.6)

The binary relation $\cong$ is an **equivalence relation** on strictly partially ordered sets. That is, let $(X, <_X)$, $(Y, <_Y)$ and $(Z, <_Z)$ be strictly partially ordered sets, then the binary relation $\cong$ satisfies the following properties:

- (a) **reflexive**: for all X , $X \cong X$;
- (b) **symmetric**: for all X and Y , if $X \cong Y$ then $Y \cong X$;
- (c) **transitive**: for all X , Y and Z , if $X \cong Y$ and $Y \cong Z$ then $X \cong Z$.

Order-Isomorphism

Definition (p. 168)

An **order-theoretic invariant** is a property of an ordered set preserved under order-isomorphism.

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Order-Embedding

Motivation

We also need to define when an ordered set fits inside another while preserving its order structure entirely.

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Definition (p. 167)

Let $<_X$ and $<_Y$ be strict partial orders on the sets X and Y , respectively.

A function $f : X \longrightarrow Y$ is an **order-embedding** of X into Y (or of $(X, <_X)$ into $(Y, <_Y)$) if f is **one-one** and, for all $x_1, x_2 \in X$,

$$x_1 <_X x_2 \quad \text{if and only if} \quad f(x_1) <_Y f(x_2).$$

Order-Embedding

Exercise (informal)

Let $<$ be the usual order on $\mathbb{N}$ and $\mathbb{R}$. Show an order-embedding of $(\mathbb{N}, <)$ into $(\mathbb{R}, <)$

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Let $<$ be the usual order on $\mathbb{N}$ and $\mathbb{R}$. Show an order-embedding of $(\mathbb{N}, <)$ into $(\mathbb{R}, <)$

Solution

Let $f : \mathbb{N} \longrightarrow \mathbb{R}$ be the function $n \longmapsto n$. We can show that the function f is one-one and that it preserves the order, that is, for all $m, n \in \mathbb{N}$,

$$m < n \quad \text{if and only if} \quad f(m) < f(n).$$

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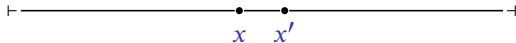
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Representation of Linearly Ordered Sets

*“If $<_X$ is the order on X , and the elements x, x' of X are related by $x <_x x'$, we picture x to the **left** of x' . [...] Of course, we must not read too much into the picture. The picture is very suggestive of the usual real line, but for instance there might not be any elements of X between x and x' in the picture above; and indeed there might be no order-embedding from X into $\mathbb{R}$ with its usual order.” (p. 167)*



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Maximum and Minimum Elements

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Maximum and Minimum Elements

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Let X be a weakly linearly ordered set by $\leq_X$ and let $a, b \in X$.

- (a) The element b is the **maximum** (or **greatest**) element of X , if $x \leq_X b$ for all $x \in X$.

Maximum and Minimum Elements

Definitions (p. 168)

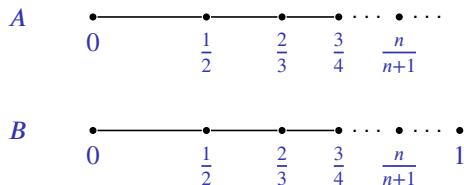
Let X be a weakly linearly ordered set by $\leq_X$ and let $a, b \in X$.

- (a) The element b is the **maximum** (or **greatest**) element of X , if $x \leq_X b$ for all $x \in X$.
- (b) The element a is the **minimum** (or **least**) element of X , if $a \leq_X x$ for all $x \in X$.

Maximum and Minimum Elements

Example (informal, p. 168)

Let $A = \left\{ 1 - \frac{1}{n+1} : n \in \mathbb{N} \right\}$ and $B = A \cup \{1\}$, subsets of $\mathbb{Q}$, with the usual order $\leq$.



The set A has minimum element, 0 , but no maximum element.

The set B has minimum element, 0 , and maximum element, 1 .

Maximum and Minimum Elements

Remark

In the next theorem we shall see that the maximum and the minimum are order-theoretic invariants.

Maximum and Minimum Elements

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Theorem 7.1

Let $\leq_X$ and $\leq_Y$ be strict partial orders on the sets X and Y , respectively.

Let $f : X \longrightarrow Y$ be an order-isomorphism from $(X, \leq_X)$ to $(Y, \leq_Y)$.

- (a) If X has a maximum element then so does Y .
- (b) If X has a minimum element then so does Y .

Maximum and Minimum Elements

Example (informal)

Let $A = \left\{ 1 - \frac{1}{n+1} : n \in \mathbb{N} \right\}$ and $B = A \cup \{1\}$, subsets of $\mathbb{Q}$, with the usual order $\leq$.

From Theorem 7.1 the ordered sets $(A, \leq)$ and $(B, \leq)$ are not order-isomorphic because the set B has maximum element, but the set A does not.

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Derek Goldrei (1996). Classic Set Theory. A Guided Independent Study. Chapman & Hall (cit. on p. 2).